1 Introduction

2 Notation

We will use $\mathbb{1}()$ to denote the indicator function (a.k.a. the truth function). $\mathbb{1}(S)$ equals 1 if S is true, and 0 if S is false. For example, $\mathbb{1}(x \geq 0)$ equals 0 if $x < 0$ and 1 otherwise.

In this paper, random variables will be indicated by underlining. For example, $P(\underline{x} > x)$ will denote the probability that the random variable $\underline{x}$ is greater than x.

bnet will be used as an abbreviation of Bayesian network.

3 Drought AA bnet

Standardized Precipitation Index (SPI) ranges from -4 (least rain) to 4 (most rain).

I_D = **Impact** (i.e., **severity**) of drought is defined as minus SPI (higher rain, higher SPI, less severe drought)

$P(I_D)$ = likelihood, probability that a drought of severity I_D will occur in the near future (weeks to months), based on forecasts.

μ = ensemble member index

M = number of ensemble members, 51 for $ECMWF$ (European Centre for Medium-Range Weather Forecasts)

$$\begin{aligned} Risk_D &= E[\underline{I}_D] = \frac{1}{M}\sum_{\mu=1}^{M} I_{D,\mu} && (1) \\ &= \sum_{I_D} P(I_D) I_D && (2) \end{aligned}$$

A **risk matrix** is a 4×4 matrix with X axis equal to impact (I) and Y axis equal to likelihood P, with I binned into four values and $P(I)$ binned into 4 values. The matrix elements equal to some arbitrary function $f(I, P)$ defined as risk. Usually, $f(I, P) = IP$ but not necessarily. I find risk matrices arbitrary and not very useful. The Wikipedia article on risk matrices (Ref.[1]) agrees.

I^*_D = impact threshold for drought (e.g., $I^*_D = -SPI = 1.5$)

$$P(\underline{I}_D \geq I^*_D) = \frac{1}{M}\sum_{\mu=1}^{M} \mathbb{1}(I_{D,\mu} \geq I^*_D) \quad (3)$$

P^*_D = probability threshold for drought
C_D = cost of drought
L_D = loss of drought

$$P_D^* = \frac{C_D}{L_D} \tag{4}$$

T_D = trigger for drought

$$T_D = \mathbb{1}\left[P(\underline{I}_D \geq I_D^*) > P_D^*\right] \tag{5}$$

Let N^D be the number of times (seasons) in which predicted and observed droughts were compared.

$N_{y|x}^D$ = number of events in which $x \in \{0,1\}$ droughts occurred and $y \in \{0,1\}$ droughts were predicted

For example, $N_{1|0}^D$ is the number of events in which no drought actually occurred but it was predicted nonetheless (False Alarm)

For clarity, will omit D superscript from N^D below.

$N_{0|0}$ = **True Negatives** (TN)
$N_{1|1}$ = **True Positives** (TP)
$N_{0|1}$ = **False Positives** (FP)
$N_{1|0}$ = **False Negatives** (FN)

$$N_{|0} = N_{0|0} + N_{1|0}, \quad N_{|1} = N_{0|1} + N_{1|1} \tag{6}$$

$$N = N_{|0} + N_{|1} \tag{7}$$

$$P(1|1) = \frac{N_{1|1}}{N_{|1}}, \quad P(1|0) = \frac{N_{1|0}}{N_{|0}}, \quad P(0|1) = \frac{N_{0|1}}{N_{|1}}, \quad P(0|0) = \frac{N_{0|0}}{N_{|0}} \tag{8}$$

$$\begin{tikzcd}
 & \underline{X}_1, \underline{X}_2, \ldots \arrow[d] \\
P(\underline{I}_D > I_D^*) \arrow[d] & P_D^* = C_D/L_D \arrow[dl] \\
\underline{T}_D \arrow[r] & \underline{T}_D^{pred}
\end{tikzcd}$$

Figure 1: AA bnet

$\underline{X}_1, \underline{X}_2, \ldots$ represents local road density, population density, historical events

A Appendix: Functions of $N_{y|x}$

$$\textbf{Hit Rate } (\text{HR})^1 = P(1|1) = \frac{N_{1|1}}{N_{|1}} \tag{9}$$

$$\textbf{False Alarm Rate } (\text{FA Rate})^2 = P(1|0) = \frac{N_{1|0}}{N_{|0}} \tag{10}$$

$$\textbf{Miss Rate } (\text{MR}) = P(0|1) = \frac{N_{0|1}}{N_{|1}} \tag{11}$$

$$\textbf{Correct Negatives } (\text{CN})^3 = P(0|0) = \frac{N_{0|0}}{N_{|0}} \tag{12}$$

References

[1] Wikipedia. Risk matrix. https://en.wikipedia.org/wiki/Risk_matrix.

[1] HR is sometimes called Precision

[2] Meteorologists also use an FA Ratio instead of FA Rate. **FA Ratio**$= \frac{N_{1|0}}{N_{1|0}+N_{1|1}} \leq 1$

[3] CN is sometimes called the **Specificity**